

EXPERIMENT 6

AIM:

To analyze daily website traffic data using time series techniques to identify trends, seasonality, and anomalies.

CODE:

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose

data = pd.read_csv("website_traffic_timeseries.csv", parse_dates=["Date"],
index_col="Date")

data.interpolate(method="linear", inplace=True)

print(data.head())

print(data.info())

print(data.describe())

from statsmodels.tsa.seasonal import seasonal_decompose

decomposition = seasonal_decompose(
    data["Page_Views"],
    model="additive",
```

```
    period=7
)

decomposition.plot()

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Moving Average
data["Page_Views_MA"] = data["Page_Views"].rolling(window=30).mean()

data[["Page_Views", "Page_Views_MA"]].plot(figsize=(12, 5))
plt.title("Page Views with Moving Average")
plt.xlabel("Date")
plt.ylabel("Page Views")
plt.show()

# Anomaly Detection using Z-score
data["Page_Views_Z"] = (
    (data["Page_Views"] - data["Page_Views"].mean())
    / data["Page_Views"].std()
)

anomalies = data[data["Page_Views_Z"].abs() > 3]
```

```
print("Anomalies:")  
print(anomalies[["Page_Views", "Page_Views_Z"]])  
  
plt.figure(figsize=(12, 5))  
plt.plot(data.index, data["Page_Views"], label="Page Views")  
plt.scatter(  
    anomalies.index,  
    anomalies["Page_Views"],  
    color="red",  
    label="Anomalies"  
)  
plt.title("Website Traffic Anomaly Detection")  
plt.xlabel("Date")  
plt.ylabel("Page Views")  
plt.legend()  
plt.show()
```

OUTPUT:

Page_Views Unique_Visitors Bounce_Rate

Date

2025-01-01	2074	1280	48.91
2025-01-02	2326	1306	44.98
2025-01-03	2999	1678	45.09
2025-01-04	3185	1863	45.93
2025-01-05	2586	1555	50.24

<class 'pandas.core.frame.DataFrame'>

DatetimeIndex: 365 entries, 2025-01-01 to 2025-12-31

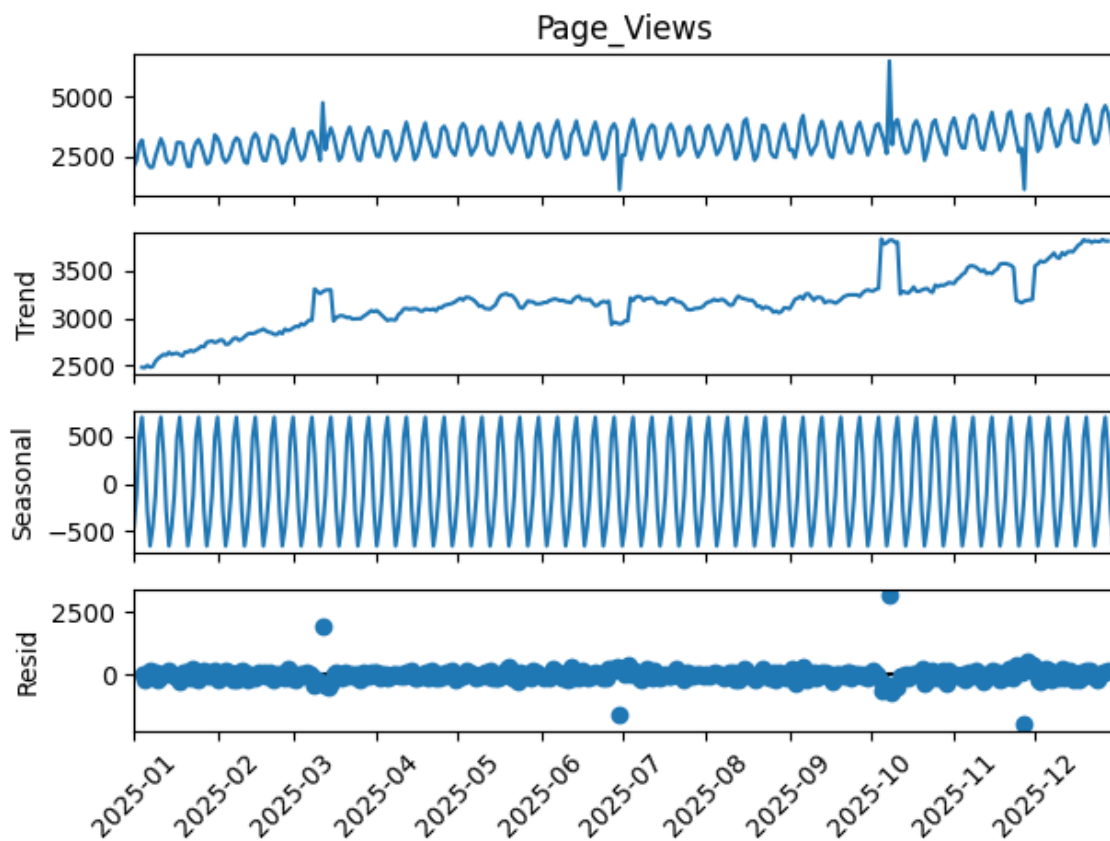
Data columns (total 3 columns):

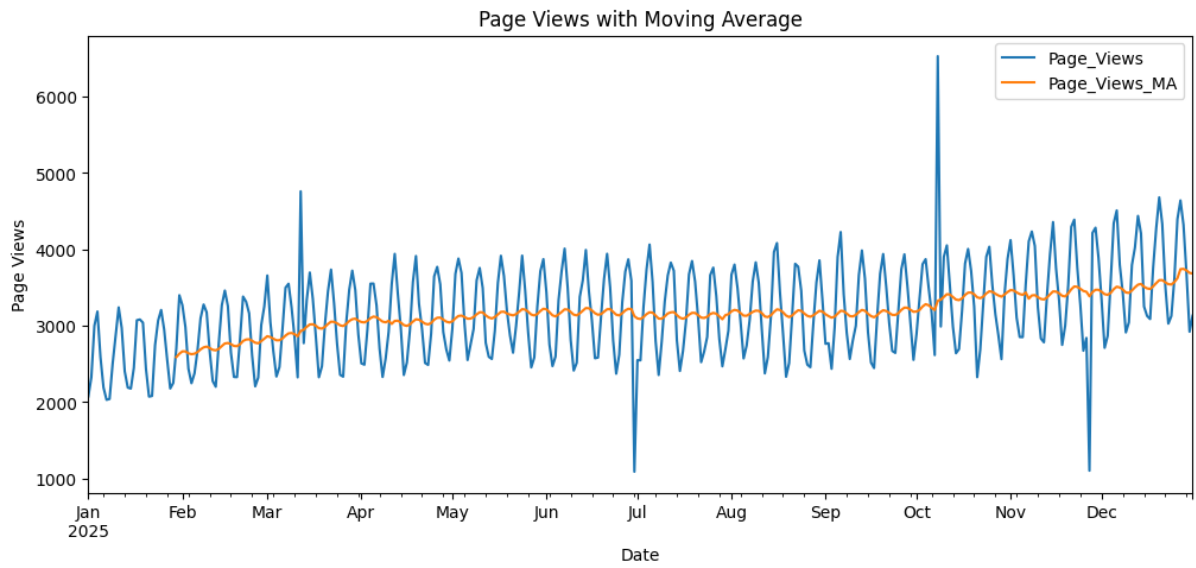
#	Column	Non-Null Count	Dtype
0	Page_Views	365 non-null	int64
1	Unique_Visitors	365 non-null	int64
2	Bounce_Rate	365 non-null	float64

dtypes: float64(1), int64(2)

memory usage: 11.4 KB

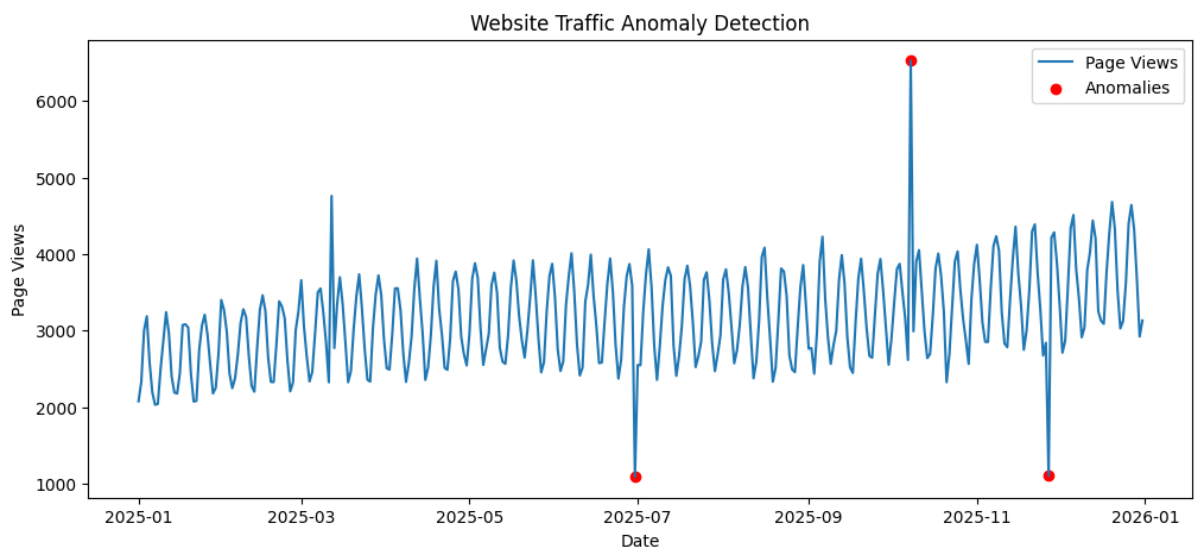
	Page_Views	Unique_Visitors	Bounce_Rate
count	365.000000	365.000000	365.000000
mean	3166.630137	1830.115068	48.036904
std	623.902005	371.898955	3.479127
min	1087.000000	617.000000	37.530000
25%	2686.000000	1550.000000	45.310000
50%	3105.000000	1797.000000	48.240000
75%	3614.000000	2083.000000	50.680000
max	6529.000000	3764.000000	60.550000





Anomalies:

Date	Page_Views	Page_Views_Z
2025-06-30	1087	-3.333264
2025-10-08	6529	5.389260
2025-11-27	1101	-3.310825



RESULT:

The website traffic data was successfully analyzed, and trends, seasonal patterns, moving averages, and anomalies were identified.